NUTRI: AI-Powered Personalized Nutrition Platform

https://github.com/rsmva-oo1/For_Hogwarts

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Project: Private AI System Submission

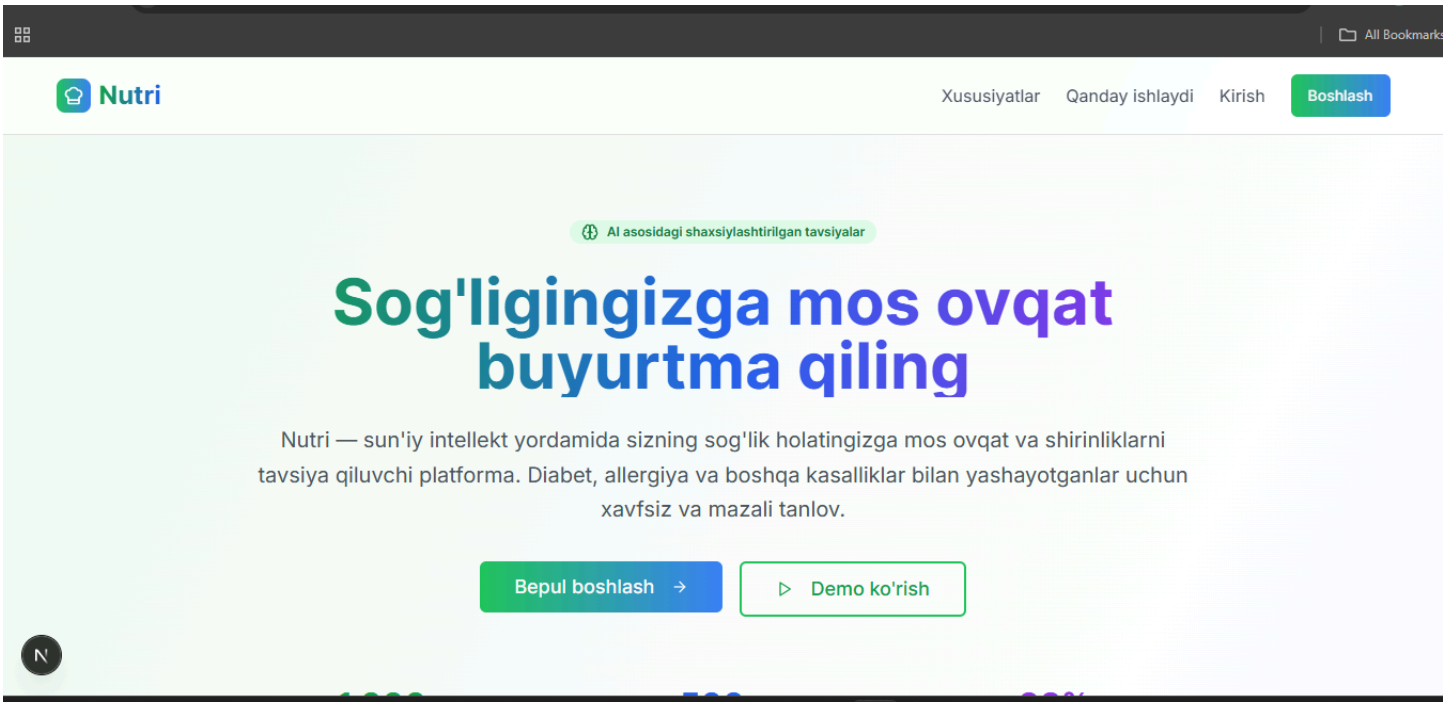
1. Project Vision

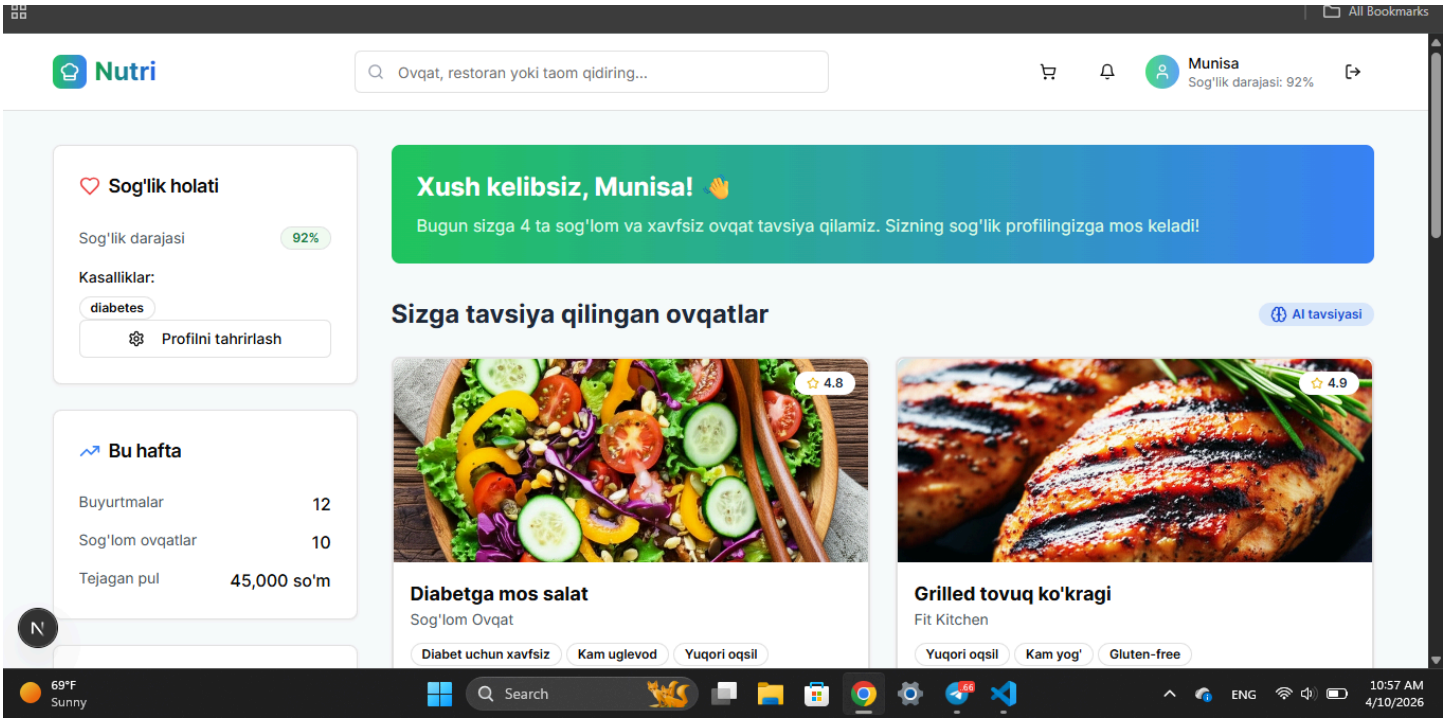
Nutri is a cutting-edge platform designed to simplify healthy eating through Artificial Intelligence. It focuses on users with specific dietary needs (e.g., diabetics, allergy sufferers) and provides them with a personalized ecosystem to discover, analyze, and order meals that are safe for their health.

2. User Interface & Experience (UI/UX)

The interface is designed with a "Health-First" philosophy. We used a **Sage Green** and **Soft White** color palette to evoke a sense of freshness and reliability.

- **Dashboard:** A clean, card-based layout that gives users a quick overview of their dietary goals.
- **AI Integration:** A seamless chat interface where users can consult with an AI Nutritionist in real-time.
- **Navigation:** A minimalist sidebar for quick access to various sections like "Home", "Meals", and "AI Consultant".





3. Technical Architecture

The application is built using a modern Full-Stack architecture to ensure speed, security, and scalability.

- **Frontend: Next.js 14** (App Router). I chose Next.js for its excellent Server-Side Rendering (SSR) capabilities, which improve SEO and performance.
- **Styling: Tailwind CSS**. This allowed for rapid UI development and a fully responsive design that works on mobile and desktop.
- **AI Engine: Groq / Llama-3-70b**. We integrated a high-speed LLM to process user queries and provide medical-grade dietary suggestions.
- **Icons: Lucide-React** for a consistent and modern visual language.

4. How the AI Integration Works

The core value of Nutri is its AI-driven advice.

1. **Prompt Engineering:** The AI is "primed" with a system prompt that instructs it to act as a professional nutritionist.
2. **API Communication:** When a user types a query (e.g., "What should I eat for breakfast as a diabetic?"), the frontend sends a request to the Next.js API route.
3. **Security:** The API request is processed on the server-side where the **GROQ_API_KEY** is securely stored in environment variables (.env). The key is never exposed to the client's browser.

5. Challenges Faced & Solutions

- **Challenge 1: Navigation Logic.** I encountered difficulties while integrating the AI chat into the dynamic sidebar menu.

    - **Solution:** I redesigned the navigation flow to prioritize the AI Assistant as a central feature on the Dashboard, making it more accessible to users.
- **Challenge 2: UI Hydration.** Ensuring that the complex UI components rendered correctly without hydration errors in Next.js.
    - **Solution:** I optimized the component structure and used useEffect hooks to handle client-side specific rendering.

6. Future Roadmap

- **Smart Meal Ordering:** Direct integration with restaurant APIs to order recommended meals.
- **Health Tracker:** A feature to track blood sugar levels and correlate them with the meals ordered through Nutri.
- **Multi-language Support:** Expanding the AI's capability to understand and respond in Uzbek and other regional languages fluently.

7. Conclusion

Nutri proves that AI can be more than just a chatbot; it can be a vital tool for improving public health. By combining a beautiful UI with powerful LLMs, Nutri makes healthy eating accessible to everyone, regardless of their medical complexity.